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Landslide susceptibility mapping using logistic regression model (a case study in Badulla District, Sri Lanka)

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Abstract

The landslide is a universal term covering a wide variety of mass movements and processes involved in downward movement of masses of rock, earth or debris under the influence of gravity. Landslides are among the natural disasters that are often experienced in Sri Lanka. Approximately 20,000 km² (30.7%) of the land area of the country is highly susceptible to landslides. With the increasing demand for development and expansion of human settlements, landslides have become a major concern in the mountainous regions of the country. Therefore, identification of landslide potential associated with the terrain is important for ensuring the sustainability of developments while minimizing the possible disasters due to landslides. The study was focused on landslide susceptibility mapping in Badulla District using logistic regression model. Slope, aspect, lithology, land cover, distance from the rivers and roads were selected as the causative factors for the analysis. According to the study, 20.5% area of the district is highly and extremely highly susceptible to the occurrence of landslides, 39.3% area is moderately susceptible to landslide occurrences and 40.2% area is low susceptible to the occurrence of landslides.

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1. Introduction

Landslide is a universal term covering a wide variety of mass movements and processes involved in downward movement of masses of rock, earth or debris under the influence of gravity. The phenomena described as landslide is not limited either to the land or to sliding, but commonly addresses almost all varieties of mass movements on the slopes, including debris falls, rock falls, topples and debris flows that involve little or no true sliding [1].

Landslides are among the natural disasters that are frequently experienced in Sri Lanka. Badulla, Nuwara Eliya, Ratnapura, Kegalle, Kandy, Matale, Kalutara Districts which are located in the central hills and Matara, Galle and Hambanthota Districts in the southern hills are identified as the landslide prone areas in Sri Lanka. Among those, approximately 20,000 km² (30.7%) of land area is highly susceptible to landslides [2]. With the increasing demand for development and expansion of human settlements, landslides have become a major concern in the mountainous regions of the country. Slope failures leading to significant loss of life and property have become a major threat to the social, economic, and environmental aspects of Sri Lanka. Therefore, identification of landslide susceptible areas associated with the terrain is important for ensuring the sustainability of developments while minimizing the possible disasters due to landslides. The spatial probability of landslide event itself can be identified through landslide susceptibility mapping.

Landslide susceptibility can be defined as the tendency for a landslide to be generated in a specific area in the future; this can be measured from the correlation between determining factors together with the spatial distribution of the movements [3]. The aim of landslide susceptibility mapping is to identify places of landslide occurrence over a region on the basis of a set of internal causative factors. This is specifically known as Landslide Susceptibility Zonation (LSZ), which can formally be defined as the division of land surface into near-homogeneous zones and then ranking these according to the degrees of actual or potential hazard due to landslides.

Among the wide range of statistical methods proposed in the assessment of landslide susceptibility, Logistic Regression (LR) analysis has proven to be one of the most reliable approaches [4,5,6,7&8]. Logistic regression analysis relates the probability of landslide occurrence (having values from 0 to 1) to the “logit” u (where $-\infty < u < \infty$ for higher odds of non-occurrence and $0 < u < \infty$ for higher odds of occurrence). In logistic regression analysis, the logit “ u ” is assumed as a linear combination of independent variables and the formulae is given as follows,

$$P_r = e^u / (1 + e^u) \quad (1)$$

Where p is the output of the model which is the probability of a landslide occurrence, and u is the independent variable which is a linear combination of the contributing factors (for example, the slope, geology, land cover, etc.). $\beta_1, \beta_2, \beta_3$, etc. are corresponding coefficients to each of the respective contributing factors which indicate their contribution to landslide susceptibility.

$$u = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \quad (2)$$

1.1. Objectives

Main objectives of the study was,

- i. To develop a spatial database for future landslide analysis
- ii. To develop a Landslide Susceptibility Map using Logistic Regression Model
- iii. To provide an improved decision support tool for hazard managers and planners

1.2. Study Area

The study was focused on landslide susceptibility mapping in Badulla District (Fig. 1) using logistic regression model. The district covers about 2871 sq. km of land area with altitude varying from 50 to 2530 m above sea level. Most of the areas in the district are densely populated residential areas and consists mostly of mountainous terrain with deep valleys. The mountain has experienced several landslides and huge human and social, economic losses in the past. The district plays a very significant role in the country's economy. In the district most of the land area is mountainous, with the growth of population, there is an increasing demand for the lands and utilization of land on hill

slopes is unavoidable. Therefore, mapping these landslides prone areas are significant to identify the distribution of hazard potential associate with those mountains for planning future development in this region.

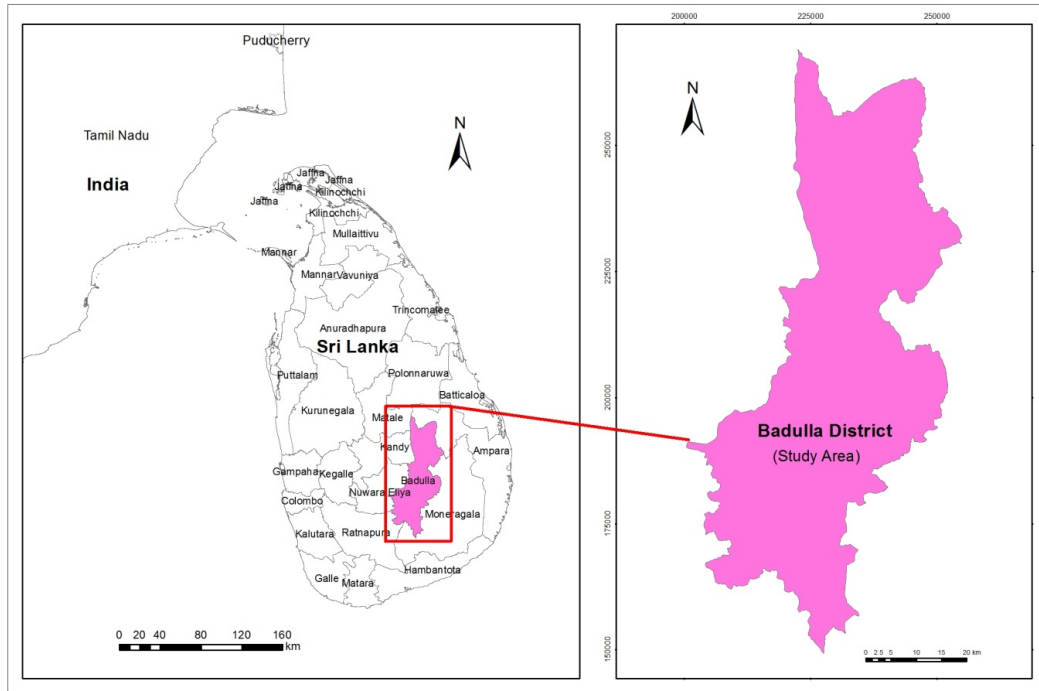
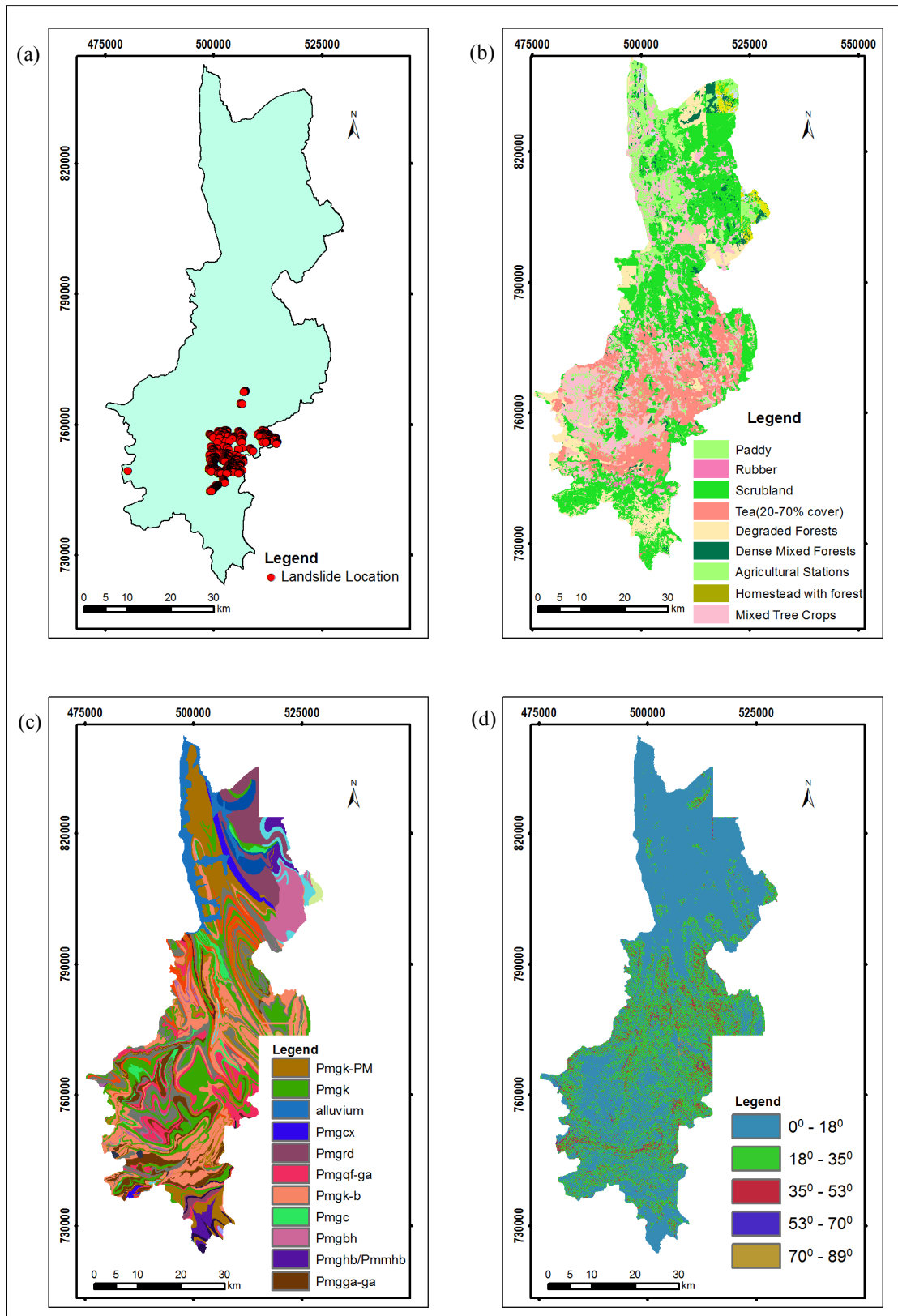


Fig. 1. Map showing the study area

2. Data acquisition and pre processing

In order to detect the landslide-related factors, some prior knowledge and literature should be prepared. Landslide related factors include three aspects such as topographic factors water-related factors and human being activity factor are applied and slope range, aspect, distance from rivers, distance from roads, land use, lithology (Fig. 2) were selected as the landslide causative factors in the present study. Lithology data was obtained from geological maps published by Geological Survey and Mines Bureau, Sri Lanka. All other data which were used to derive the other factors were obtained from the Survey Department of Sri Lanka. Landslide inventory of the selected area was obtained from the National Building Research Organization, Sri Lanka. The approach mentioned is based on the assumption that past environmental conditions at the time of landslides can be keys to evaluate the potential sites for landslides in the future. Then, all the landslide-related factors were created and store in the spatial database by using Arc GIS software in order to have the same extents, spatial resolutions, projections, etc.



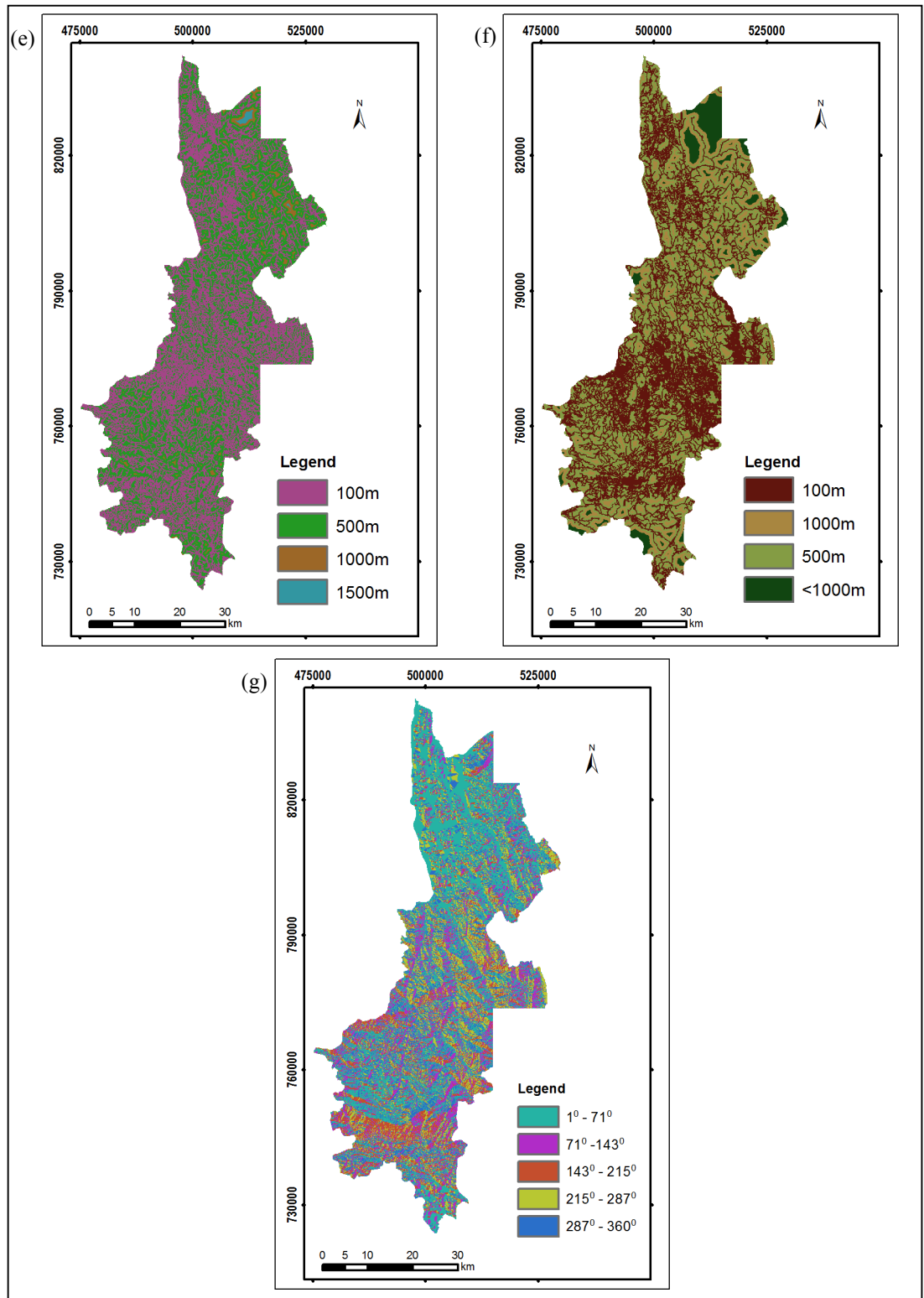


Fig. 2. Maps showing the landslide inventory and the causative factors of the study - (a) landslide inventory; (b) land use; (c) lithology; (d) slope range; (e) distance from river; (f) distance from road; (g) aspect

3. Methodology

The working procedure is shown in Fig. 3. This is followed by an assessment of the landslide susceptibility using the relationship between landslides and causative factors, and the subsequent validation of results. A key feature of this method is that the probability of landslide occurrence will be comparable with observed landslides. All the necessary data layers were derived as necessary and after exporting data into suitable exchange formats, weights of each category of the each factor were determined to account the contribution of each factor to the occurrence of landslides. Then the effective factors were selected for susceptibility analysis and the factors that corresponding the distance from river and distance from roads were eliminated due to its zero value in weight calculation and all other factors were get into account. After recombining the coefficients, as seen in Eq. (2), the possibility of occurrence of landslides was finally computed and susceptibility map was produced. Landslide susceptibility in the resulted map was categorized into four classes utilizing natural breaks classification as, low susceptibility, moderate susceptibility, high susceptibility and extremely high susceptibility. Then validation of the results done and the validation procedure was based on a comparison between the results provided by the LR and an external dataset (not used in the training stage).

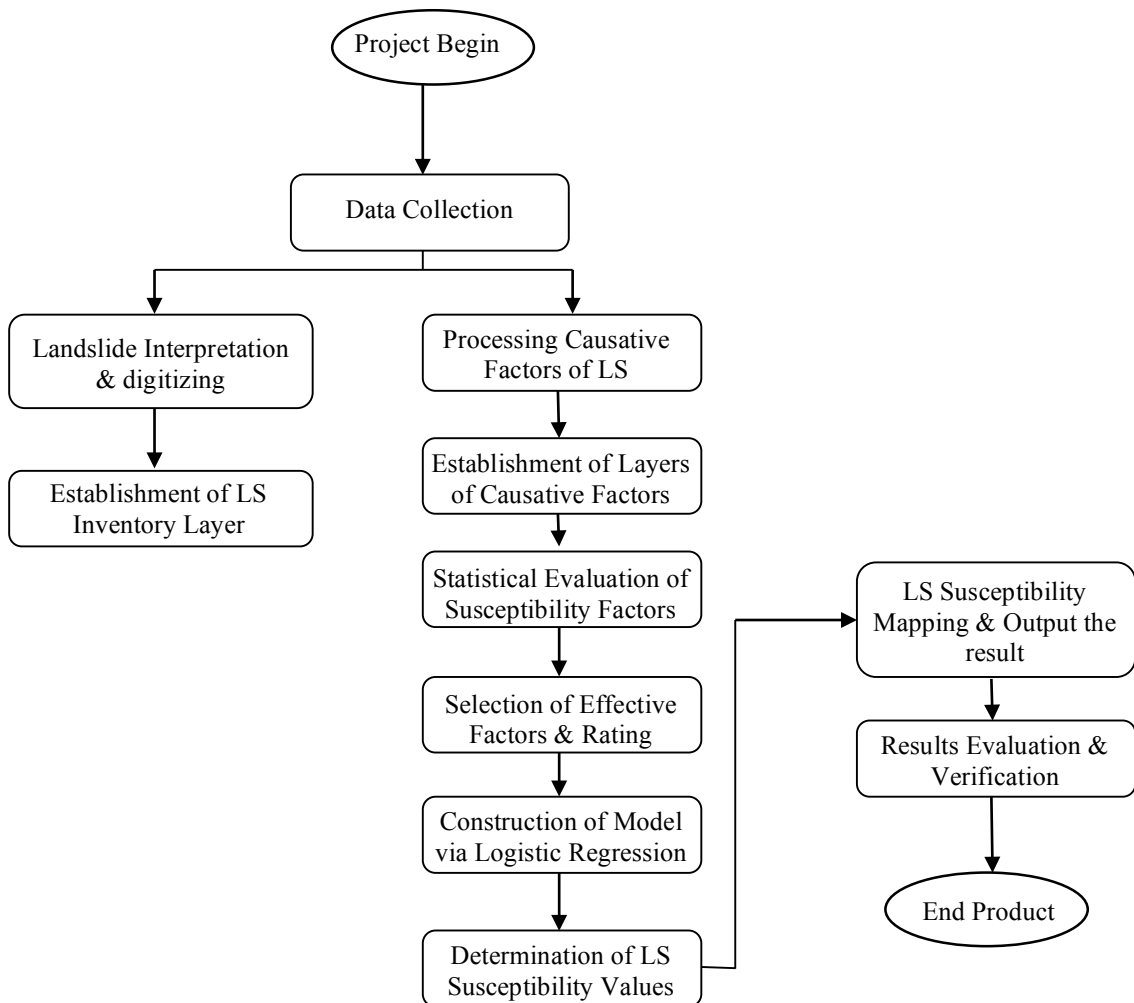


Fig. 3. Figure showing the methodology of the study

4. Results and discussion

The relative importance of independent variables to the occurrence of landslides can be expressed by the regression coefficient (Table 1). The results revealed that the land use factor has the highest coefficient and a large percentage of the weighting in the study area and it is the most influencing factor to the occurrence of landslides of the area. Slope, lithology and aspect also make a considerable contribution to the occurrence of landslides. The contribution of the distance from river and distance from road to the occurrence of landslides was negligible in this area.

Table 1. Coefficient of logistic regression for each factor

Factor	Coefficient
Land use	3.55
Slope	1.36
Lithology	0.64
Aspect	0.40

Resulted landslide susceptibility map of the study area is shown in fig. 4 and according to the established classification; approximately 20.5% area is classified as high and extremely high susceptible zone. Moderate susceptible zone covers 39.3% of the total area and it is only 40.2% of total area included as a low susceptible zone.

The landslide susceptibility analysis results were verified using existing landslide location. To validate the accuracy of the landslide susceptibility map, observed landslide and susceptibility map was compared. The verification results showed satisfactory agreement between the susceptibility map and the existing data on landslide location. It showed that a high percentage of the observed landslides points (76%) were located in high and extremely high susceptibility areas, that covers the lower percentage of the study area and the model seems to be reliable in Badulla district.

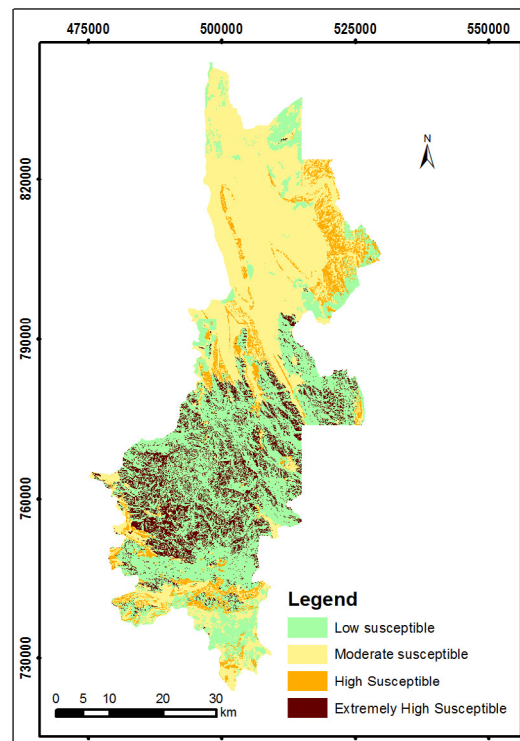


Fig. 4. Resulted landslide susceptibility map for the study area

5. Conclusions

It was concluded that, logistic regression model clearly highlights the interrelation existing between the occurrence of landslides and instability factors. The relative importance of independent variables can be expressed by the regression coefficient.

It was found out that, land use factor most strongly related to slope failure occurrences than other factors. In particular, factors such as “distance from river” and “distance to roads” show zero weight and, it can be obtained due to the negligible influence on landslide occurrences in this study.

Validation results of the landslide susceptibility analysis showed that most of the observed landslide points were located in high and very high susceptibility categories. Hence, it may be inferred that the map correlates well with existing field conditions. The model seems to be reliable to the selected areas.

The inability to find a well distributed landslide inventory map was the main limitation of this study.

6. Recommendations

The accuracy and prediction ability of the landslide susceptibility map could offer us crucial information for city planning, infrastructure construction and agriculture developments in the future or in other area with similar conditions.

The landslide susceptibility map could be used for general purposes, like for the decision making of slope remedial measure, regional planning and hazard mitigation policy.

Conversely, the quality of the susceptibility map can be further improved by incorporating more factors.

Further, any change in the natural environment by human interference, such as implementation of development projects, deforestation, may change the existing landslide susceptibility of the area. Hence, such maps should be updated periodically.

If the factors relevant to vulnerability of population, buildings, villages and other property are available, risk analysis also can be done and incorporating the result with temporal factors, landslide hazard zonation map also can be developed.

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